Bishow Lamsal

Agentic AI | Generative AI | Machine Learning Engineering

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PROFESSIONAL SUMMARY

AI/ML Software Engineer with 4+ years of experience designing, training, and deploying production-grade AI systems across agentic automation, NLP, computer vision, and generative AI. Experienced in end-to-end delivery, from training large language models and designing scalable REST and gRPC APIs to architecting multi-agent agentic workflows, RAG pipelines, and deploying real-time vision systems on edge devices. Delivered high-impact solutions for global clients including the United Nations FAO and US-based financial sector clients, integrating systems with developer and project management platforms such as Jira, and Microsoft Teams.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript
- **AI / ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, YOLO, Pandas, NumPy
- **LLMs & Agentic AI:** OpenAI GPT, Anthropic Claude, LLaMA, Phi, Qwen, LangChain, LangGraph, RAG, MCP, Playwright
- **Backend & APIs:** FastAPI, Flask, gRPC, REST APIs
- **Databases & Vector Stores:** MongoDB, PostgreSQL, Redis, Qdrant, MEM0
- **Deployment & MLOps:** Docker, AWS, CI/CD Pipelines, MLflow
- **Monitoring & Observability:** Grafana, Prometheus, Loki, cAdvisor, TensorBoard, Weights & Biases
- **Edge AI & GPU:** CUDA, TensorRT, cuDNN, NVIDIA Jetson Xavier

WORK EXPERIENCE

Machine Learning Engineer · Vertex Special Technology — *Jan 2024 – Present*

Core AI/ML software engineer building production agentic systems for international clients across QA automation, financial intelligence, and UN procurement.

- Architected and led development of AI Stellar, an agentic AI QA automation platform built using Playwright and LLM-driven UI agents that generates hundreds of reusable, structured test scenarios from a single natural language prompt, with native support for developer workflow integrations including Jira stories, achieving 5–10× faster test execution versus manual workflows.
- Built a gRPC-based conversational interface that interprets user intent and dynamically routes between UI automation and test case generation using context-aware multi-agent decision making.
- Integrated Model Context Protocol (MCP) to enable seamless tool connectivity across agent workflows, supporting flexible and extensible tool use within the agentic pipeline.
- Designed a DOM-caching strategy that eliminates redundant LLM calls for repeated or modified test cases, significantly cutting automation cost and inference latency.
- Engineered a layered memory architecture with dedicated short-term and long-term memory between agent nodes, enabling coherent multi-step reasoning across complex automation workflows.
- Implemented Human-in-the-Loop (HITL) checkpoints that detect low-confidence inputs, pause agent execution, prompt user clarification, and rerun the pipeline with validated context.

- Built a full CI/CD pipeline with real-time observability using Grafana, Prometheus, Loki, and cAdvisor across all deployed agents.
- Integrated multi-model fallback across leading LLM providers for high availability and reliability, alongside prompt versioning, model versioning, and granular token usage and cost tracking per pipeline run.
- Delivered an end-to-end agentic procurement system for the United Nations FAO, automating the full RFQ and ITB workflow including document parsing, data extraction, structured analysis, spreadsheet generation, and LPC report creation, achieving above 95% end-to-end system accuracy, deployed within Microsoft Teams via Power Automate and LangGraph, enabling a single operator to replace multiple staff.
- Delivered ScAInit, a financial document intelligence system for a US-based client combining layout detection models, OCR engines, and fine-tuned Vision Language Models (Phi, Qwen), achieving 95% layout accuracy on bank statements and 90% table structure recognition across complex borderless and multi-span tables.

Machine Learning Engineer · Bottle Technology *Sep 2022 – Sep 2023*

Designed and deployed real-time computer vision systems on live city infrastructure across Kathmandu and Lalitpur for traffic analytics and crowd management.

- Developed and deployed a real-time traffic analytics system across 4 cameras at 4 city locations, processing live footage at 20 FPS on NVIDIA Jetson Xavier edge devices.
- Implemented vehicle detection and tracking using YOLO and DeepSORT with geometric and motion-based logic to identify one-way violations and illegal highway stops.
- Optimized model inference via TensorRT, significantly improving GPU utilization and sustaining 20 FPS throughput on constrained edge hardware.
- Analyzed 100+ hours of recorded footage to deliver actionable traffic insights directly supporting urban planning decisions for Kathmandu and Lalitpur Metropolitan authorities.
- Conducted real-time crowd management analysis at BNB Hospital and ticket counters, tracking large volumes of individuals to support operational capacity planning.
- Trained specialized OCR models for Devanagari script license plate recognition and Nepali ID card extraction, enabling accurate reading of Nepali-language documents.
- Built and deployed end-to-end ML pipelines from model development to cloud integration using Docker, Jenkins, and AWS.

Machine Learning Trainee · Bottle Technology *Jun 2022 – Aug 2022*

- Implemented classical image processing algorithms from scratch based on research papers, with and without OpenCV.
- Annotated training datasets, fine-tuned detection models, and tracked experiments using TensorBoard and Weights & Biases.
- Benchmarked multiple YOLO versions for real-time object detection, comparing performance across speed and accuracy trade-offs.

KEY PROJECTS

Healthcare Data Extraction · *Cloud-Based PDF Health Records Pipeline · US Client*

- Delivered a scalable pipeline processing approximately 1TB of psychology-related health records (split PDFs) for a US-based client, performing redundancy detection, region identification, and structured extraction of key-value pairs and tabular data into MongoDB with a fully rule-based approach that kept all patient data on-premise, eliminating LLM exposure and significantly reducing operational costs.
- Delivered a Streamlit UI enabling rapid patient data retrieval across thousands of psychology documents in minutes with zero LLM dependency, ensuring full compliance with US healthcare data sensitivity requirements.

Face Detection, Recognition & Data Extraction · *Computer Vision Identity Pipeline*

- Built a face similarity detection system with document classification and structured data extraction using a multi-engine OCR pipeline (PaddleOCR, TrOCR, EasyOCR) for maximum accuracy across varied document formats.
- Deployed on Nepali Cloud to ensure complete data sovereignty, keeping all sensitive identity data within national infrastructure.

Advanced E-Video-KYC · *Real-Time Liveness Authentication*

- Developed a real-time liveness detection system recognizing facial movements (blink, mouth open, head turns), integrated with OCR-based document extraction and voice assistant support.

News Recommendation System · *Hybrid Engine with Multi-View Attention*

- Built a hybrid recommendation engine on the MIND dataset combining content-based filtering, collaborative filtering, and popularity signals using Word2Vec embeddings and Multi-View Attention.

Real-Time Work Surveillance · *AI-Powered Employee Activity Monitoring*

- Developed an employee activity monitoring system with real-time face detection and recognition via vector databases, tracking seated hours, mobile usage, and behavioral patterns with web-based visualization.

EDUCATION

Bachelor of Engineering in Computer Science *Aug 2016 – Dec 2020*

Visvesvaraya Technological University, Bangalore, India · CGPA: 7.53 / 10

CERTIFICATIONS & TRAINING

- Hackathon Mentor, Hack4SafeFood (UN FAO-aligned food security hackathon)
- Data Science Training, 4 months, Deerwalk Training Center
- Web Development Internship, 7 weeks, Rbits Technology
- Active Kaggle competitor with hands-on experience across real-world datasets from Kaggle and UCI Repository